Aaryan Kandiah

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Applied AI Engineer | Agentic AI · RAG · Data Pipelines | Available May - August 2027

PROFESSIONAL SUMMARY

AI engineer and NUS Master of Computing (AI) student with a year of engineering experience at Univers, building RAG-based agentic platforms and on-premises LLM deployments for clients, plus an internal AI monitoring service. 2× 1st-place winner and 1st runner-up at AI competitions, building multi-agent systems, retrieval pipelines and MCP integrations. High-agency builder who takes prototypes through to deployed, working systems.

WORK EXPERIENCE

Univers, AI Innovation Engineer Intern | Python, Java, TypeScript, React, Docker May 2025 – Apr 2026

- Engineered full-stack AI HVAC Agentic Platform for internal GTM and engineering teams, featuring RAG architecture with VectorDB, context-aware recommendations, role-based reasoning, and AI-powered sales enablement tools increasing GTM team productivity by ~25%.
- Optimised Java backend and React/TypeScript frontend energy-saving dashboards for 8 clients; debugged critical backend API endpoints, implemented real-time chatbot streaming, natural language data queries, and interactive visualisations, reducing load time by 20%.
- Deployed 2 on-premises AI HVAC solutions on dedicated hardware with local LLM hosting, configuring end-to-end AI dashboards and backend/frontend services via Dockerized deployments on Linux servers, enabling AI capabilities with zero cloud dependency.
- Built internal AI monitoring service to detect anomalous data in real-time across client-facing dashboards, with automated notifications to engineering team, reducing incident response time by ~40%.

EDUCATION

National University of Singapore (NUS) Aug 2026 – Dec 2027 (Expected)

- Master of Computing (Artificial Intelligence)
- Relevant Modules: (1) *Neural Networks and Deep Learning* [Linear Algebra, Transformer Architecture] (2) *Distributed Systems* [RPC/RMI, Multithreading, Consistency & Replication] (3) *Reinforcement Learning and Sequential Decision Making* [Markov Decision Processes, Rational Decision Making]

Nanyang Technological University (NTU) Aug 2022 – Dec 2025

- Bachelor of Engineering (Electrical and Electronic Engineering)
- Specialisation: Data Analysis and Machine Learning
- Honours (Distinction) CGPA: 4.36 / 5.00
- Relevant Modules: (1) *Data Structures and Algorithms* (2) *Machine Learning Design & Application* [Large Language Models, Transformer Architecture, Attention Mechanism] (3) *Intelligent Systems Design* [Deep Learning, Computer Vision, Statistical Pattern Recognition]

Virtual Training and Learning & Development

- Amazon Web Services (AWS) Certification: *AI Practitioner*
- Google Certification: *Advanced Data Analytics Certificate*

ACADEMIC PROJECTS / COMPETITION PROJECT / RESEARCH EXPERIENCE

AI Workflow Competition, Echelon 2026 – 1st Place | FastAPI, LangChain, Python, Next.js Jun 2026

Project: BOLDR SignalDesk – Approval-First AI Customer Intelligence Workflow for SME Support

- Secured $16,000 in prizes for architecting a human-in-the-loop AI platform automating SME customer support, spanning evidence-grounded reply drafting and marketing-intel generation from resolved queries.
- Developed multi-source evidence retrieval pipeline (FastAPI backend) unifying policy, product, and pricing documents into a single structured knowledge base for real-time reply drafting.
- Directed reply drafting and customer service resolutions via LangChain-based GLM-5.1 orchestration layer with deterministic fallback logic, sustaining ~96% answerability accuracy.
- Designed self-improving knowledge loop that converts human-resolved gaps into reusable knowledge-base entries, reducing repeat escalations and projecting ~24 hours/week of support time saved at pilot scale.

Agent Forge AI Hackathon (Z.ai and GMI Cloud) – 1st Place | FastAPI, LangChain, Python, React Apr 2026

Project: ADapt – AI-Powered Ad Localization Platform for Southeast Asian Markets

- Awarded 1st place for building an end-to-end AI pipeline, localising ads across 6 SEA countries, generating culturally adapted strategies, multilingual copy, and localised imagery from text or image input.
- Architected a multi-model orchestration system (FastAPI + React) chaining GPT-5.4 (vision/text input), GLM-5.1 (strategy & copywriting), and Gemini 3.1 Flash (image generation) via GMI Cloud APIs.
- Implemented image-to-image editing via Gemini 3.1 Flash multi-turn contents format, enabling surgical edits (e.g., product swaps) that preserve original ad composition.
- Integrated Publer for direct TikTok publishing, enabling end-to-end workflow from ad ingestion to live social deployment in under 3 minutes.

NTU-EEE Module: Final Year Project | PyTorch, Scikit-learn, OpenCV, Python Jan 2025 – Dec 2025

Module Project: Multi-Label Chest X-ray Abnormality Detection Using Deep Learning and Explainable AI

- Developed multi-label classification model using modified DenseNet-121 in PyTorch to detect 14 thoracic abnormalities (e.g., pneumonia, cardiomegaly) from the NIH Chest X-ray dataset (~112k images).
- Implemented custom weighted loss function to address class imbalance, reducing false negatives by 15%.
- Improved model generalisation through data augmentation and batch normalisation, achieving average AUC score of 0.845, matching performance of several state-of-the-art CXR studies.
- Integrated Grad-CAM Explainable AI for clinical interpretability of model predictions on X-ray regions.

Qoder x Alibaba Cloud Hackathon – 1st Runner-Up | TypeScript, React, Node.js, Python Aug 2026

Project: Trace – Local-First AI Decision-Memory Platform

- Constructed a local-first AI platform that autonomously captures browser research and organises evolving decisions into Git-inspired threads, branches and commits, mitigating the need to re-research past decisions.
- Engineered an autonomous evidence-routing pipeline that classifies and files research within 10-25 seconds and reconciles compatible conclusions across branches at 95%+ confidence with no human approval step.
- Integrated Trace with Qoder via a read-only MCP server exposing 5 decision-retrieval tools (search, trace history, current answer, constraints, prior regrets), for direct in-IDE access.
- Developed a React decision-history canvas, a privacy-aware Chrome capture extension, and a native SwiftUI menu-bar app, forming a full local-first system spanning capture, storage, and UI.

Independent Project | Elasticsearch, MCP, ES|QL, Python Feb 2026

Project: Research Orchestration System – AI Literature Review & Claim Verification Agent

- Built a multi-agent AI system automating literature reviews and claim verification across ~200 papers (~5,000 chunks indexed in Elasticsearch), orchestrating three agents in an iterative feedback loop.
- Designed a Research Agent pipeline that decomposes queries into sub-questions, scopes the corpus via ES|QL, and retrieves evidence through hybrid keyword and semantic search.
- Implemented a Peer Review Agent that verifies citations, audits confidence tags claim-by-claim, and returns actionable revision feedback, enforcing a structured PASS/REVISION_NEEDED verdict loop.
- Developed a custom indexing pipeline (PDF parsing, chunking, 384-dim embeddings) and exposed the system via React web app, Slack bot, REST API, and MCP endpoint.

NTU-EEE Module: Computer Vision | Scikit-learn, TensorFlow, Python Aug 2024 – Dec 2024

Module Project: Enhancement of Brain Tumour Classification Using Dimensionality Reduction Techniques

- Benchmarked PCA, LDA, PCA+LDA, and Autoencoders against six classifiers on ~4,000 MRI images, reducing 16,384-pixel dimensions to 1,413 (91% reduction) while preserving 95% of dataset variance.
- Built a full preprocessing pipeline (resizing, augmentation, grayscaling, standard scaling) and evaluated all 24 model/reduction combinations on accuracy, precision, and recall.
- Achieved up to 94.35% accuracy with PCA despite a 91% dimensionality reduction, outperforming LDA (~68%) due to its linear class-separability assumption.

Undergraduate Research (URECA) | PyTorch, Python Aug 2023 – May 2024

Module Project: Fairness Analysis for New Interpretable Machine Learning Architecture (TTnet)

- Benchmarked Truth Table Net (TTnet), an inherently interpretable CNN architecture, against 4 models (XGBoost, Decision Tree, CNN, FairCORELS V2) across 6 fairness metrics on COMPAS and Adult datasets, concluding TTnet fairest on 5 of 6 metrics while also leading on accuracy (84% vs. 78–82% on Adult).
- Implemented a fairness-aware training method using proxy-metric loss functions, differentiable approximations that enable gradient descent on otherwise non-differentiable fairness constraints.

SKILLS

- **Programming Languages:** Python, Java, TypeScript/JavaScript, SQL
- **Agentic AI & LLM:** Multi-agent orchestration, RAG (Hybrid search, Embeddings), MCP, LangChain, LLM evaluation, LLM API integration
- **Data & Backend:** FastAPI, REST APIs, Data pipelines, Elasticsearch/ES|QL, PostgreSQL, Redis, Vector DBs, Node.js
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Linux, Git, GitHub Actions, CI/CD
- **ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas
- **Dev Tools:** Claude Code, Antigravity, Cursor, Codex
- **Languages:** Proficient in English; conversant in Hindi and French